

Explorative application of discrete Bayesian networks as surrogate models for energy systems analysis

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HIGHLIGHTS

- We present methods for building, validating, and using BBNs as surrogate models.
- With average relative error below 2 %, the BBN model is an adequate model surrogate.
- Benefits: interactive analysis, intuitive uncertainty propagation, goal-driven analysis.

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ABSTRACT

This work investigates data-based discrete Bayesian Belief Networks (BBNs) as surrogate energy system models for result analysis and interactive analyses, such as what-if analyses. A simplified version of the Swiss TIMES (STEM) model, referred to as STEM-lite, is used for demonstration. A method to optimize the BBN model is devised, based on performance metrics related to the accuracy of the BBN predictions, calculated over data records unseen by the BBN in the training phase. Further validation of the BBN on a set of seven scenarios yielded an average relative error below 2 %, suggesting adequate performance as surrogate model. The application of the surrogate BBN model is demonstrated to highlight its benefits, which include enabling interactive analysis (supported by the visualization of key variables, their relationships and interactions), fast and intuitive uncertainty propagation, and support for goal-driven analysis (backward reasoning from outcomes to the inputs that produce these outcomes). The surrogate BBN presented here was developed to elaborate the methods for constructing, validating, and using BBN models for energy systems analysis and to demonstrate the benefits of such a model; at this stage, this model is not intended for energy systems and economics policy discussions. For practical applications, future work is needed to reduce the number of data records to construct the BBN, to introduce the option to treat the time dependence of the input variables, and to allow for larger BBN models (involving more variables) that reflect the increasing complexity of energy systems.

1. Introduction

Energy systems models provide comprehensive and detailed representation of the energy system sectors, from production to energy conversion and consumption, and are typically used to identify energy system pathways that achieve one or more objectives under different policies, subject to technical, resource and other constraints [1]. Today, a primary application of energy system models is the achievement of

sustainability in the context of climate change. The models are built with detailed technology specifications and often employ linear programming techniques to minimize the overall cost of the energy provision. The generated outputs include estimates of technology capacity and utilization, marginal commodity prices, and emissions across the energy system. Some of the analytical strengths of energy systems models include a consistent framework for specifying techno-economic performance, efficient goal-seeking within complex systems, and insights into

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various energy futures aligned with policy objectives. The energy systems analysis space is typically multi-dimensional, spanning over energy technologies, commodities (e.g. electricity, gas, etc.), sectors (e.g., buildings, electricity sector, transport, etc.) as well as addressing multiple time and space scales. For this reason, the resulting mathematical model couples time, space and technology among the dimensions as well as within each dimension. The extensive scope of energy system models has led to increasing complexity as various approaches and features are developed to enhance the realism of internal model dynamics [2,3].

Due to the complexity of the problem and the large number of interdependent variables (e.g. policy parameters, technology costs, consumer needs and uses, sector coupling), the analyses can be computationally expensive (i.e. hours on standard computers). Also, interpreting the results in terms of the interactions among the variables and the model outputs requires expertise in energy economics and energy systems analysis, as well as familiarity with the underlying modelling frameworks. The treatment and communication of uncertainties and the transparency of the connection among assumptions, models and the produced insights, e.g. in terms of the drivers of the results themselves and of the derived policy recommendations, are recurrently identified as areas in need of improvement [4–6].

When the computational demands of models are high, uncertainty propagation and analysis, sensitivity analyses, as well as prompt what-if analyses can be carried out by introducing surrogate models, computationally efficient approximations of more complex models. The use of simplified models to increase computational efficiency is established in energy systems analysis: heuristics and machine learning approaches are primarily used [7–10]. Dominant applications of machine learning address forecasting of prices and demands [7,8,11]. Applications of machine learning for energy system modelling has also received attention, as recent reviews point out [9,10], mostly to develop models from field data patterns, e.g. from smart meters and smart building sensors. Applications for surrogate modelling of energy systems models have used traditional approaches such as neural networks and support vector machines, with the primary objective of obtaining fast-running model approximations [12,13]. While offering an adequate solution to the computational speed, these machine learning approaches do not address the needs for transparent communication of results, of the associated uncertainty and policy recommendations.

The development of surrogate energy system models to support analysis and communication of the produced insights and the associated uncertainties is the subject of the present paper. The transparency of the surrogate model is emphasized in light of the growing demand for explainability in machine learning models, especially when these are used to inform policies with important societal implications [14]. In particular, this paper addresses graphical, probabilistic models for an intuitive visualization of the large-scale energy systems model variables and of their relationships: Bayesian Belief Network (BBN). The visualization capabilities of BBNs supports the explainability of the developed model [15]. The BBN network structure represents causality relationships across variables, showing the chain of causal dependence from model inputs to results, therefore potentially allowing to explore the drivers of the results and of the recommended policies. BBN's interactive framework lends itself to stakeholder involvement in the analysis of the results and implications. BBNs are typically used to represent and analyze uncertain knowledge through graph structures, i.e. Directed Acyclic Graphs (DAGs). The underlying relationships are estimated through statistical, probabilistic and computational methods, making Bayesian networks a combination of graph theory, probability theory, computer science, and statistics [16]. The BBN modelling flexibility is demonstrated by the large variety of application areas, e.g. natural and social sciences, engineering, finance, medical diagnosis [17]. For data-rich domains, BBN models can be constructed from the empirical data; in other cases, the BBN is constructed from expert judgment, e.g. for causal models of rare events. The foundation of BBNs in the Bayesian probabilistic framework also supports a formal combination of empirical

and expert judgment data.

The research objective of the paper is to investigate and demonstrate the attractiveness of BBNs for surrogate modelling, in particular focusing on uses of interest for energy systems analysis, representation and propagation of uncertainty, goal-driven and what-if analyses. BBNs have been widely applied in the energy sector, as presented in recent review papers [18–20]. Data-driven BBN applications typically generate predictions concerning the energy market [21–24], electrical demand (e.g. from smart city data) [25], and renewable resource predictions (e.g. wind data series) [18]. Many studies exploit the BBN's capability to support reasoning under uncertainty, generally for representation of causal relationships on uncertain variables. Examples include identification and quantification of influences on prediction of carbon price [26], energy security [27], future demand for the water, energy, food nexus [28–30], reduction of CO₂ emissions [31–33], effectiveness of policy and project implementation [34–36,37], optimal siting identification [38–40]. Very recently, BBNs were used to investigate interdependencies and related uncertainties among technologies within decarbonization pathways, as it emerges from ensembles of energy system model runs [41].

The energy model utilized in this work to explore the BBN application is a simplified version of the Swiss TIMES energy model, developed in earlier work by [42], referred to as STEM-lite. The main simplifications introduced in STEM-lite relate to reducing: the scope of the Swiss TIMES model to the electricity sector only, the number of yearly time-slices (from 288 to 32), categorized into four seasons, with separate consideration for working days and weekends in each season; the number of periods in which the 2020–2050 time horizon is divided (three periods of 10 years each). Like STEM, STEM-lite is implemented in General Algebraic Modelling System (GAMS), an algebraic modelling language commonly used in energy system analysis [3].

The paper is organized as follows. The next section presents the basics of the BBN modelling tool, focusing on the BBN type adopted for the present demonstration: discrete BBNs. The section also presents the methodology to develop the BBN as a surrogate model, comprising three steps: STEM-lite data generation and discretization; identification of the optimal BBN with respect to specified performance metrics; validation of the optimal BBN on seven scenarios related to energy systems policy exploration. Section 3 demonstrates potential uses of the BBN as a surrogate energy systems model, building on two scenario cases. Discussion and conclusions are given in Sections 4 and 5, respectively.

2. Materials and methods

2.1. Bayesian belief networks

In the qualitative BBN representation, the DAG, nodes denote random variables and edges indicate the presence of a relationship among these variables [16,17]. Basic concepts on BBNs are given in Appendix A. The direction of the edges indicates the direction of influence, with parent nodes influencing child nodes. The BBN node relationships are quantified by conditional distributions. For discrete BBNs, i.e. BBNs where the nodes represent discrete random variables or a discretized representation of continuous variables, the relationships take the form of conditional probability distributions. For continuous BBNs, relationships are represented by conditional probability density functions. Hybrid BBNs combine discrete and continuous variables.

The present work adopts discrete BBNs: this is a common modelling strategy which, compared to continuous BBNs, allows using simpler learning and inference algorithms [43–45]. The *GeNIe* software package [46] was used for the BBN development and visualization.

A fundamental assumption for BBN models is that the joint probability of the variables $X_1, \dots, X_N$ can be factorized according to:

$$P(X_1, \dots, X_N) = \prod_{i=1}^N P(X_i | pa(X_i)) \quad (1)$$

where $pa(X_i)$ is the set of parents of X_i as represented in the DAG.

Depending on the problem at hand and the data availability, different methods are used to build a BBN, blending expert knowledge and data in different ways and proportions, both to develop the BBN structure and to estimate the distribution parameters. The development of the BBN structure (i.e. the DAG) typically involves data-mining algorithms to identify the causal relationships among the variables. These algorithms are data intensive, so that BBN structure learning is often aided by expert knowledge in the pre-selection of the most significant variables and links, and in the identification of the direction of influence. In many applications dealing with the causation of rare events, the DAG is fully derived by expert knowledge, exploiting a variety of knowledge elicitation techniques [17]. The quantitative relationships, the conditional distributions, can also be quantified solely using data, with distribution-fitting algorithms, combining data and judgment within a Bayesian framework, or directly via expert elicitation [47].

For the development of the surrogate model in the present paper, data-rich methods for BBN development are of interest. Indeed, STEM-lite's computation time (about 3 s per run on a standard computer) allows generation of many data records, up to 100,000 as presented in Section 2.3. In particular, methods for BBN structure learning have been extensively studied in the literature and are typically distinguished as: score-based, constraint-based and hybrid [48,49]. Score-based methods evaluate different BBN models in the space and select the ones that optimize an information criterion, typically based on goodness-of-fit to the data. Constraint-based algorithms, on the other hand, perform a sequence of independence tests and construct a directed acyclic graph (DAG) based on the learned conditional independence relations. Hybrid algorithms combine elements from the two approaches.

For the present work, two structure learning algorithms were tested, both implemented in the *GeNIe* software package: Greedy Thick Thinning (GTT) [50], and the Bayesian Search [51] algorithm, which are among the most used algorithms of the score-based type (note comparative study [48] shows that score-based algorithms are often faster and more accurate than constraint-based and hybrid). Preliminary tests resulted in the choice of the GTT algorithm, which consistently exhibited faster performances (up to more than two orders of magnitude in computing times) and comparable accuracy.

For the estimation of the BBN discrete conditional probability distributions, *GeNIe* implements the Expectation-Maximization (EM) algorithm, which iteratively updates an initial distribution to maximize the likelihood of the available data [46,52].

2.2. BBN model development - overview

The process devised to develop the surrogate BBN model comprises three steps:

1. Generation and discretization of the STEM-lite data for learning and validation
2. Optimization of the BBN structure and parameters
3. Scenario validation of the BBN as surrogate model

The first step (Section 2.3) involves generation of STEM-lite input-output records and subsequent data discretization, for the records to be processed by the discrete BBN learning algorithm. Discretization of continuous variables must be done carefully, to avoid losing statistical accuracy and deriving inappropriate conclusions [44]. The approaches to variable discretization can be classified in: 1) manual, i.e. based on expert knowledge; 2) supervised, i.e. informed on the output variable, typically via optimization algorithms; 3) unsupervised, i.e. based on the distribution of the variable itself to be discretized [43,44]. The most

widely applied approaches in BBN applications are manual and unsupervised, the latter most frequently adopting equal-width or equal frequency bins [43,44]. While supervised approaches are frequently adopted in the general computer science literature, their use for BBN modelling is still limited, due to both knowledge gaps and lack of implementation in most commercial software [44]. Manual discretization, also known as expert discretization, involves the user manually selecting the thresholds for dividing the continuous data into discrete bins, on the basis of physical significance, theoretical knowledge, or the user's expertise in the specific problem domain [44]. Anticipating the discussion in Section 2.3, for the present application, the input variables (Table 1) were discretized with equal-width bins, while the other modelling variables were discretized manually, based on their distribution from the input-output records.

The second step (Section 2.4) determines optimal modelling choices affecting the BBN structure and the values of the BBN parameters (the Conditional Probability Distributions, CPDs): the number of discretization bins for each node; the maximum allowed number of parents; the data set size (and its partition in training and validation data). As presented in Section 2.4, the options for each of the three choices determine 36 combinations ($3 \times 3 \times 4$, for the respective choice), therefore corresponding to 36 possible BBNs. Three evaluation metrics are used to determine the optimal BBN, two metrics relate to the BBN prediction accuracy, one relates to the BBN information content. Prediction accuracy is embedded into the *GeNIe* software and is calculated via confusion matrices based on 20 % of the input-output records, unseen by the BBN in the training phase.

For each node subject to the validation check, the BBN prediction is the node state with the largest probability, thus the prediction accuracy

Table 1

STEM-lite inputs: fixed and uncertain. Uncertain variables correspond to the inputs in the input-output records used to develop the surrogate BBN model.

Economy and population ⁽¹⁾	2030	2040	2050
Gross Domestic Product (BCHF)	695.6	846.4	981.6
Population (million)	9.4	10	10.3
Resource potentials for electricity production⁽²⁾	Maximum value allowed		
Solar (TWh/yr)	50		
Wind (TWh/yr)	4.3		
Biomass (TWh/yr)	8.3		
Geothermal (TWh/yr)	4.3		
Maximum Gas imports (TWh/yr)	100 ⁽³⁾		
Maximum electricity net import (TWh/yr)	100 ⁽³⁾		
Maximum nuclear power installed capacity (GW)	20 ⁽³⁾		
Maximum gas power plant installed capacity (GW)	20 ⁽³⁾		
Uncertain variables⁽⁴⁾	Sampling range, Uniform distribution		
Nuclear power plant investment cost [CHF/kW]	5000–20,000 ⁽⁵⁾		
Carbon tax [CHF/tCO ₂]	0–100 ⁽⁶⁾		
Wind turbine investment cost [CHF/kW]	500–4000 ⁽⁵⁾		
Electricity import price [CHF/MWh]	29.1 ⁽⁷⁾ – 300 ⁽³⁾		
Solar PV price investment cost [CHF/kW]	200–2000 ⁽⁵⁾		
Gas import price [CHF/MWh]	14.8 ⁽⁷⁾ – 100 ⁽³⁾		
Combined cycle gas turbines investment cost [CHF/kW]	500–2000 ^{(5) (8)}		

⁽¹⁾ Reference for economy and population data: [53].

⁽²⁾ Reference for resource potentials: [53], Unless otherwise stated.

⁽³⁾ Very high value compared to what achievable: implies no limit for practical purposes.

⁽⁴⁾ Each sampled value assumed as representative, constant over the 2030, 2040, 2050 time slices.

⁽⁵⁾ Ranges for investment costs adapted from [54].

⁽⁶⁾ Range based on historical values [55].

⁽⁷⁾ Range based on historical values from 2021/2022 [56].

⁽⁸⁾ Step 2 of the BBN development (Section 2.4) showed negligible effect of input variable “Combined cycle gas turbines investment cost” on the output variables of the input-output records and therefore was not included as BBN node.

metrics are based on single-state predictions. Therefore, accuracy-based validation does not address the whole BBN output of interest, i.e. the probability distributions associated to the output nodes. Also, any data record outside the most-likely discretization bin would be classified as wrongly predicted, thus penalizing predictions associated to wider distributions (see the discussion in Section 2.5 on this matter). To overcome these limitations of the accuracy-based validation, Section 2.5 presents the so-called “scenario validation”, where the node distributions of the optimal BBN are validated against STEM-lite runs on seven scenarios mimicking different policy investigation cases.

2.3. Data for BBN learning and validation (step 1)

Table 1 presents the STEM-lite inputs, used to generate the learning and validation data set, with the corresponding source. STEM-lite inputs include both fixed values, as well as uncertain variables (Table 1). The values of the uncertain variables are sampled from Uniform distributions to generate the input-output records used to develop the surrogate BBN model. The uncertain variables include the costs of electricity production technologies with the largest share in the Swiss energy mix, as well as gas and electricity import prices. The carbon tax value is explicitly considered as a representative policy instrument. Other policy instruments such as subsidies for renewable technologies can be implicitly incorporated within the costs of the associated technologies. The set of variables in Table 1 is chosen to investigate the BBN properties for simplified, yet representative case studies.

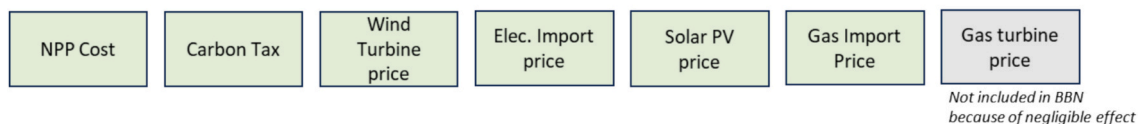
To further simplify model development for the present exploratory application, the STEM-lite uncertain input variables are modeled constant over time, i.e. over 2030–2050. In practical terms, each sampled value for the uncertain variables is applied to all three time slices 2030, 2040 and 2050 in input to STEM-lite. This simplification does not allow

modelling price or subsidy trends over time, however some representation of input variability can be obtained via their marginal probability distributions defined for the base case scenario (Section 3.1). Representation of the time dynamics is an important subject of future work.

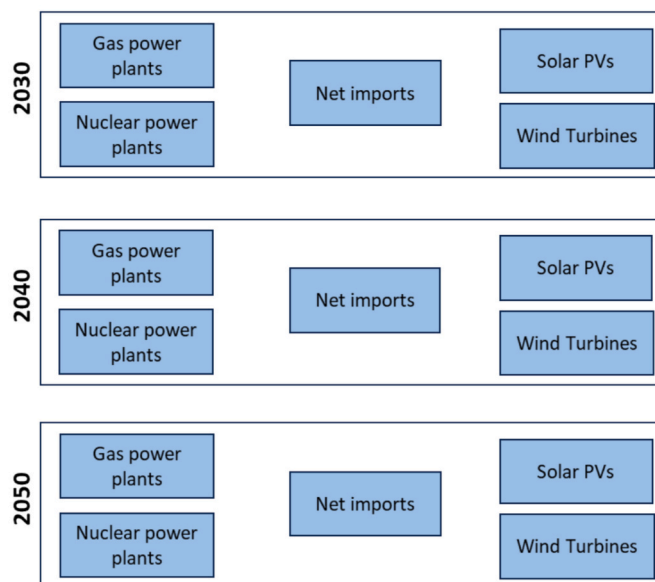
STEM-lite was run for each sampled combination of the uncertain inputs (Table 1), obtaining 100,000 input-output records (to be divided into datasets of different sizes in the BBN learning phase). The input-output records are made of each sampled combination (inputs), and the STEM-lite generated values (outputs) for the 2030, 2040 and 2050 amounts of electricity production from selected technologies (gas-fired and nuclear power plants, solar panels and wind turbines) and of the electricity import. As figures of merit for the energy configuration, the record includes the corresponding 2030, 2040, and 2050 CO₂ emissions and marginal electricity costs (also referred to as outputs). Each data record includes values of 28 variables, i.e. 7 randomly generated input values and 21 STEM-lite generated values (Fig. 1). Electricity production technologies such as hydropower and geothermal energy were not selected because of their small variability across the sampled records.

Discretization of the values in the input-output records was done differently for the input variables (uncertain variables in Table 1 and green boxes in Fig. 1) and the output variables (blue boxes in Fig. 1). The seven input variables were discretized with equal-width bins, a natural choice given their Uniform distribution, with number of bins to be determined based on the optimization procedure in Section 2.4. The other variables (electricity production shares, CO₂ emissions and marginal electricity costs in 2030, 2040, 2050) were manually discretized, based on their distribution from the 100,000 input-output records. Manual discretization was chosen because of the highly non-uniform and non-normal nature of the output distributions, see Fig. 2 for an example. The definitions and discretization intervals for all model variables are given in Appendix B, obtained via visual inspection of each

Uncertain inputs of the energy system model



Energy system (productions and imports)



Figures of merit

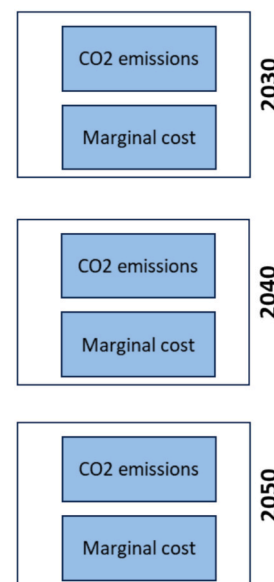


Fig. 1. Variables included in the input-output records. Inputs (green boxes): uncertain STEM-lite inputs; outputs (blue boxes): 2030, 2040, 2050 electricity production from selected technologies and corresponding 2030, 2040, 2050 CO₂ emissions and marginal electricity costs. Descriptions of the variables and units can be found in Table 1 for the uncertain inputs and in Appendix B for the STEM-lite generated quantities. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

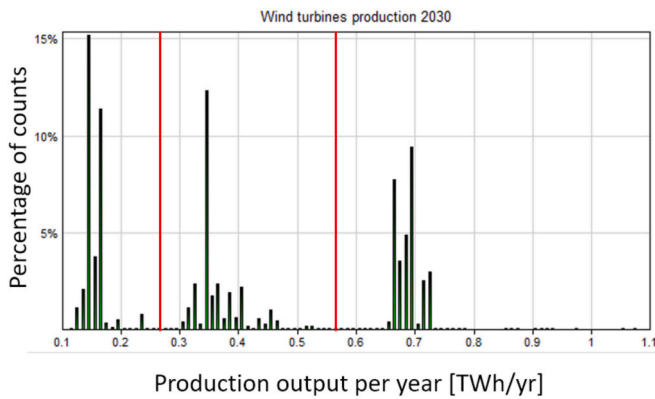


Fig. 2. Manual discretization example for STEM-lite output data: “Wind turbines production 2030”. Vertical lines identify discretization bins.

distribution, as necessary making sure that the distributions peaks would be enclosed into one bin (as shown in Fig. 2). The *GeNIe* software convention is followed for the discretized state labels: labels in growing order, s1 representing the lowest absolute value. The label also includes the lower and upper thresholds of each state bin. Only integers are used, therefore for example in Fig. 2, the second state “s2” of “Wind turbines production 2030” is named “s2.0.0”.

Indeed, manual discretization can introduce biases in the BBN response, partly because the distribution informing the discretization (e.g. in Fig. 2) is conditional on the uniform distributions assumed for the input variables. Therefore, the same discretization intervals may not be optimal for other input distributions possibly assumed when the BBN is used for other applications. In addition, the definition (number and size) of the discretization intervals can be representative of the distribution of the specific variable to discretize, but not necessarily optimal for the overall BBN response. This is why the most advanced discretization approaches address variables jointly, not individually, as an optimization problem, with objective function depending on the overall response [43,44]. As advocated in [43,44], the present paper adopts a comprehensive validation strategy both in the BBN optimization structure (Section 2.4) as well as for the final BBN structure selected for demonstration (Section 2.5).

2.4. Optimization of the BBN structure and parameters (step 2)

The development of the BBN involves modelling choices affecting its structure and the values of the BBN parameters (its CPDs): the number of discretization bins for each input node (discretization for the other nodes is done manually); the maximum allowed number of parents; the data set size (with 80 %–20 % split between learning and validation). In what follows, BBNs corresponding to each combination of choices are identified by the triplet (“# of bins”, “max parent”, data set size). The evaluation metrics used to determine the optimal BBN are forward and backward accuracy and the parameter uniformity metric.

Concerning the accuracy metrics, in forward validation, the BBN is evaluated given evidence on the input nodes, in correspondence to the states associated to the discretized input values of the input-output record selected for validation (for the reader not familiar with BBNs, Fig. 5 can be referred to as an example of the developed discrete state BBN with green boxes as inputs and blue boxes as outputs, corresponding to the energy system model variable in Fig. 1). For each node of the BBN subject to validation, the node prediction (i.e. the node state with the largest probability) is compared to the corresponding discretized output state. A node-specific accuracy is calculated for all output nodes as the fraction of the correct predictions over the validation set: forward accuracy is calculated as the average over all output nodes. In backward validation, evidence is assigned to the BBN output nodes, in correspondence of the output values of the input-output record selected for

validation. The most likely states of the input nodes are then compared to the discretized input values of the record. The backward accuracy metric is calculated based on the predictions concerning the input nodes (the most likely input values, not considering uncertainty) in the same way as the forward metric.

The BBN information content is accounted for by the parameter uniformity metric. The metric counts the number of CPDs that are still at their initial distribution (Uniform), i.e. no data was available for the corresponding parent state combination. The metric is the ratio of that number to the total number of CPDs. Low uniformity implies that a large fraction of the CPDs have been informed by data. The number of parameters of the BBNs subject to optimization varies from about 2500 for the smallest network, BBN (3,3,20k) to about 178,000 for the largest network, BBN (7, 7, 80 k), so it can very well be that the largest BBNs are overparametrized, i.e. including parameters that are not learned from data, nor subject to validation.

Fig. 3 shows the optimization metrics for the investigated combinations: # of bins, max parents and data set size, respectively ordered in the figure. Note that the number of parameters required by the BBN options increases with all the BBN optimization variables (# of bins, max parents and data set size). In particular, for the latter, the larger the data set, the larger the number of links that are added because substantiated by data.

Forward accuracy increases in all three dimensions (blue circles in Fig. 3). Larger number of bins for input variables and of linked parents correspond to higher modelling flexibility and therefore higher accuracy. This is because the amount of generated data is sufficiently large that the validation data does not exercise the BBN outside the training range, for all BBN options. With respect to the sample size, forward accuracy tends to increase, albeit very slowly for some BBN options. This is because the increasing sample size provides more information to the BBN (hence the accuracy increase), but already at 20 k, the sample size is sufficient to cover satisfactorily the data records used for test. The increasing trend in forward accuracy with the sample size is more visible for larger BBNs (right part of Fig. 3), for which the larger sample sizes still provide useful additional information.

Backward accuracy (orange triangles) decreases with the number of discretization bins for the input variables (note the steps down at the continuous lines in Fig. 3). This is the result of the fact that in the data records originated by STEM-lite, multiple input variable combinations can lead to the same output variable combination. Therefore, when evidence is set for the BBN output nodes, the accuracy score is updated only if the corresponding input data combination matches the combination of the most likely states predicted by the BBN. In all other case, the mismatch is considered as wrong prediction. The larger the number of input bins, the larger the chances that the input combination from the data would not match the combination of the most likely input states as predicted by the BBN: hence the decreasing trend. Furthermore, still from Fig. 3, backwards accuracy overall exhibits smaller values than forward accuracy: this is due to the larger uncertainty affecting the input nodes than output nodes, which again increases the chances of inaccurate predictions. These aspects show a limitation of measuring accuracy based only on the most likely node state, not considering the node distribution. Indeed, the more comprehensive validation of Section 2.5 will be performed comparing the distribution predicted by the BBN to the one originated by the validation data.

Parameter uniformity (gray squares) is strongly affected by the maximum number of allowed parents, which in turn affects the number of BBN links (and therefore parameters). For BBN options with three parents maximum, the parameter uniformity is around 0.2, i.e. 80 % of the BBN relationships are informed by the data. For the other options, parameter uniformity is above or about 0.5, which entail overparametrized BBNs with large number of parameters not informed by data. Note also from Fig. 3 that uniformity is in practice independent on the sample size, except for the larger BBN options with seven discretization bins. This means that increasing the data does not inform

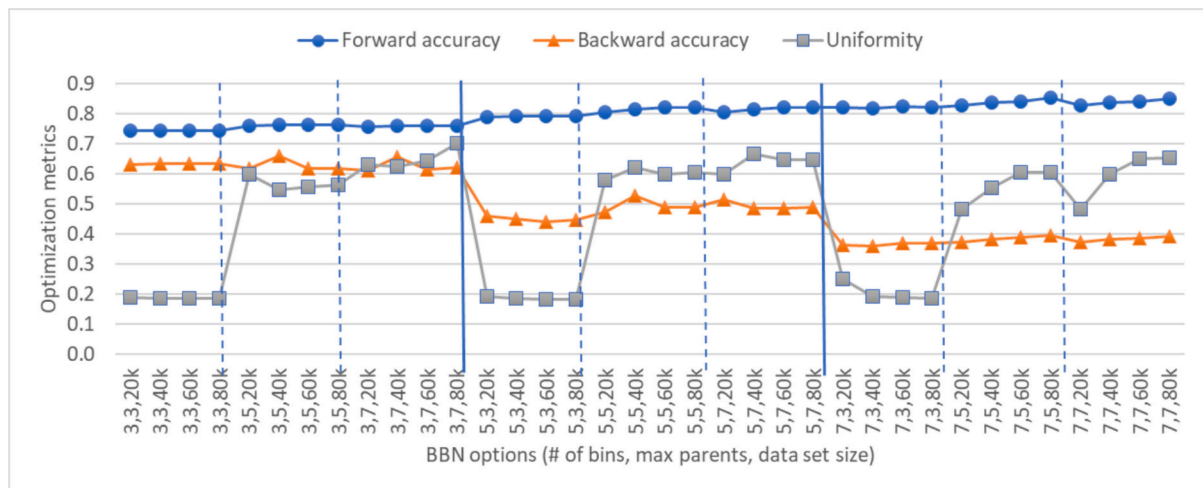


Fig. 3. Investigation of optimal BBN concerning number of input discretization bins (# of bins), maximum allowed number of parents (max parent), and size of data set. Continuous vertical lines indicate change in “# of bins”, dotted vertical lines indicate change in “max parent”.

additional BBN parameters (which would imply a decrease in the uniformity) nor establishes additional links (which could imply increase or decrease in the uniformity depending on the relationship between the additional data and the additional links). For the BBN options with seven discretization bins, uniformity decreases for the BBNs with three parents maximum, while increases for the other options.

As a result of the above considerations, the BBN structure with five discretization bins for input variables and maximum three parents allowed, BBN with (5, 3, x) coding in Fig. 3, was selected for scenario validation (Section 2.5) and, eventually, demonstration on case studies (Section 3). This choice allows for low values of the uniformity metric,

combined with high forward accuracy. Backward accuracy is mid-way between the other two BBNs structures, but, as said above, the indications from this metric should be taken with care and a more comprehensive assessment of the BBN predictions is performed in the next Section. Furthermore, the five-bin discretization is believed to offer adequate discrimination to support insights from the case studies in Section 3.

Furthermore, it is important to note that the structure learning algorithm (GTT), never included input variable “Combined cycle gas turbines investment cost” as BBN input, implying its negligible effect on the other BBN model variables. This reflects the dominant contribution

Table 2

Absolute errors of marginal distributions in scenario validation.

#	Scenario - short description	Evidence (node: state) ¹	0.5 k	1 k	5 k	10 k	20 k	80 k
1	Low emissions	CO2 Emissions 2050: s1	2.63 %	2.07 %	1.14 %	1.09 %	0.97 %	1.80 %
2	Low emissions, low renewable prices	CO2 Emissions 2050: s1 Wind turbine prices: s1	2.52 %	1.86 %	1.10 %	0.93 %	1.80 %	1.54 %
3	Low emissions, no gas	Solar PV prices: s1 CO2 Emissions 2050: s1 Gas 2030: s1 Gas 2040: s1 Gas 2050: s1	3.27 %	3.52 %	1.94 %	2.13 %	0.93 %	2.14 %
4	Low emissions, low renewable prices, no gas	CO2 Emissions 2050: s1 Wind turbine prices: s1 Solar PV prices: s1 Gas 2030: s1 Gas 2040: s1 Gas 2050: s1	3.15 %	3.63 %	1.83 %	1.64 %	1.12 %	1.04 %
5	Low emissions, low renewable prices, no imports	CO2 Emissions 2050: s1 Wind turbine prices: s1 Solar PV prices: s1 Net import 2030: s1 Net import 2040: s1 Net import 2050: s1	3.16 %	3.44 %	2.17 %	3.70 %	1.96 %	1.93 %
6	Low emissions, low renewable prices, no imports, no gas	CO2 Emissions 2050: s1 Wind turbine prices: s1 Solar PV prices: s1 Net import 2030: s1 Net import 2040: s1 Net import 2050: s1 Gas 2030: s1 Gas 2040: s1 Gas 2050: s1	3.22 %	3.44 %	1.74 %	1.77 %	1.24 %	1.04 %
7	Low emissions, high electricity and gas prices	CO2 Emissions 2050: s1 Electricity import price: s5 Gas import price: s5	2.16 %	1.71 %	1.51 %	1.15 %	1.03 %	2.25 %
Scenario average			2.87 %	2.81 %	1.63 %	1.77 %	1.29 %	1.68 %

¹ State definitions and discretization levels are reported in Appendix B.

of the cost of gas supply over the investment cost on the overall cost of electricity production from gas turbines. For this reason, the BBNs shown in this paper feature six inputs instead of seven.

2.5. Scenario validation (step 3)

While forward and backward accuracy address validation of single-state predictions, this section addresses validation of the probability distributions predicted by the BBNs, for the selected BBN configurations (5, 3, x).

Validation is carried out on seven scenarios (Table 2), hence “scenario validation”, for the two BBNs constructed from the 20 k and 80 k data sizes, with four additional BBNs built from smaller data sets, to check the effect of the data size on model performance. For each scenario, and for each BBN, the evidence corresponding to the scenario feature is set on the relevant BBN nodes (third column of Table 2). For example, for scenario 1 (“Low CO₂ emissions”), evidence for node “CO₂ Emissions 2050” is set to the lowest state, which corresponds to emissions below 2 Mt./yr (Appendix B). After evidence is set, the updated probability distributions for all other BBN variables are compared to the corresponding discretized distributions from the 100,000 input-output records, conditional on CO₂ emissions in 2050 represented by discretization bin “s1_below_2”. Fig. 4 shows the comparison for selected variables, i.e. the 2050 electricity production sources, emissions and associated costs: the distributions match satisfactorily.

Table 2 summarizes the validation results. The table reports the average of the absolute prediction error between the state probability produced by the BBN and that corresponding from the discretized data. The average is calculated over all states of all BBN nodes for which evidence is not set (i.e. nodes not mentioned in the third column of Table 2). The table shows satisfactory predictions across all scenarios, already achieved based on the 5 k data set. Note that the above error is calculated on the absolute values of the probabilities, and therefore could translate into large prediction errors for small probability values (e.g. if below 0.05).

Overall, the validation results underscore satisfactory performance for data set sizes above 5 k, as the average absolute error remains below 2 %, adequately representing a surrogate STEM-lite model, at least for

the validated scenarios. The BBN obtained from the 20 k data set, corresponding to the lowest average prediction error (Table 2) will be used for the use cases in the next Section.

3. Using the model: Scenario analysis results

The methods and results related to the development, optimization, and validation of the surrogate BBN have been presented in Section 2. This section turns to the application of the BBN as a surrogate energy systems model, demonstrating three uses:

- Representation of uncertainty in inputs and its propagation through the BBN model
- Goal-driven analyses, e.g. low CO₂ emissions, with identification of input values to meet the goals (“Backward evidence propagation” in BBN terminology)
- What-if analyses, exploring effect of changes in input values (of policy implementation) on the model outputs.

For all three uses, the speed and visualization capabilities of the surrogate model allows analysts to work interactively with the energy systems model.

Scenario analyses were carried out for eight cases, a base case scenario and the seven scenario cases used in model validation (Section 2.5). In the base case, the BBN inputs are set to represent current values and their future trends are based on realistic assumptions. For the seven scenario analysis cases, the descriptor and the evidence setting are shown in Table 2 (second and third columns). An important difference between the BBN model runs carried out for model validation (Section 2.5) and those carried out for the scenario analyses discussed in this section concerns the initial input distributions. In model validation, uniform distributions are used for these inputs (Table 1); the aim is to validate the model over the entire analysis space. In contrast, for the application of the BBN, realistic distributions are used for these inputs – these correspond to the base case distributions. Nevertheless, given the simplifications introduced, namely the reduced set of modeled variables and their time invariance, the presented results are not intended for real-world policy decisions.

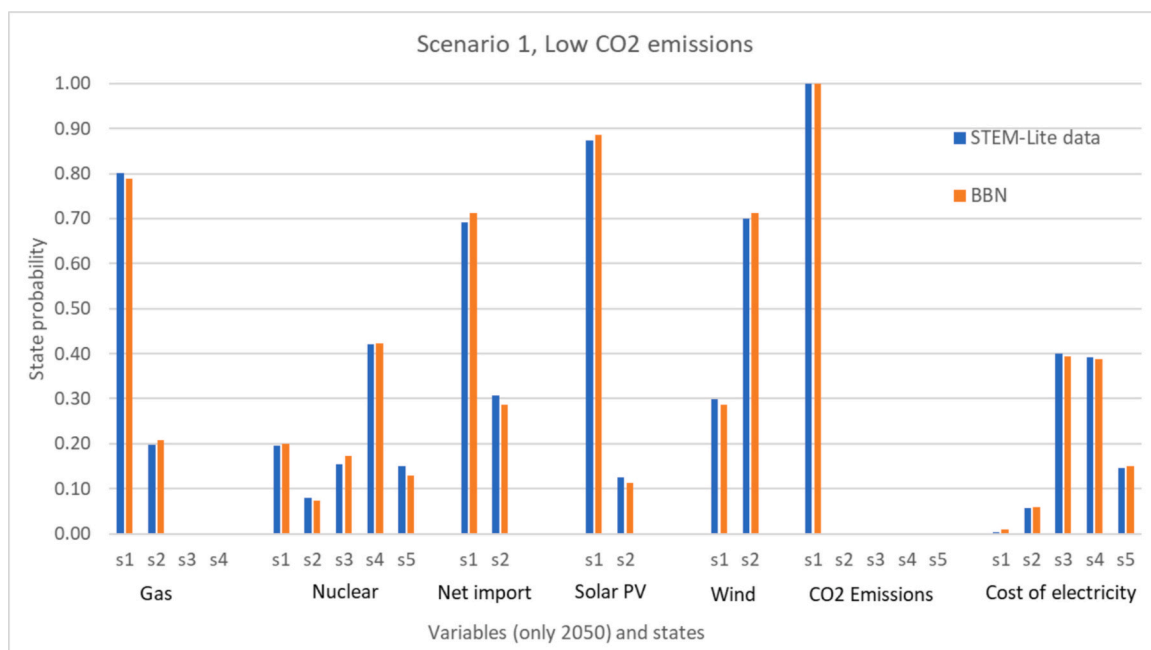


Fig. 4. Validation on scenario 1 (only 2050 variables shown). State s1 of “CO₂ Emissions 2050” is selected with certainty. STEM-Lite data is obtained from the 100,000 input-output records, conditional on “CO₂ Emissions 2050” corresponding to s1.

The BBN responses for three scenarios are discussed in detail below. These are the base case scenario and two further scenarios: scenario case 1 (Low emissions), and scenario case 2 (Low costs for renewable energies) from Table 2. The BBN responses for the remaining 5 scenarios are shown in Appendix C.

3.1. Base case scenario

The base case scenario involves assessing reference probability distributions for the six BBN inputs (green boxes in Fig. 5). The intuitive representation of the uncertainty in the inputs and of its propagation to the other model nodes is evident from Fig. 5.

Concerning the assessment of the input distributions, the mode of the marginal distributions of technology investment costs, nuclear, wind, and solar, have been set based on current values: nuclear power in the 8–11 kCHF/kW bin (state s2), wind turbines in the 1.9–2.6 kCHF/kW bin (state s3), solar PV in the 920–1280 CHF/kW bin (state s3). The distributions were defined considering that future cost reductions due to accumulated experience may be more likely than cost increases (this translates in larger probability masses for smaller values than for larger values than current). Possible subsidies for electricity production from wind and solar PVs also concur in decreasing future investment costs for these technologies.

The mode of the carbon tax is set on the 20–40 CHF/tCO₂ bin, which

correspond to CO₂ prices before the 2022 energy crises [55]. The distribution represents that larger values are considered as more likely than smaller values, given current pricing trends and long-term decarbonization goals.

Assessing electricity and gas import prices poses challenges due to uncertain developments and trends. The base case considers that the projected electricity import prices for the upcoming and the subsequent year are approximately 150 €/MWh [57] hence in the 137–191 CHF/MWh bin. The intermittency of renewable sources, weather conditions, and emerging technologies introduce significant uncertainties to electricity prices, which are reflected in the relatively large tails of the marginal distribution for the electricity import price. Concerning gas import prices, the base case considers overall trends rather than extreme scenarios. Prior to 2021, gas prices in the Dutch TTF market remained below 30 €/MWh [58], hence in the 31–48 CHF/MWh bin. While geopolitical factors still contribute to some market uncertainty, current prices have returned to pre-2021 levels.

Analyzing the generation outputs in the base case, gas power plants still play a significant role in the generation capacity due to low gas import prices and the ongoing nuclear phase-out. Indeed, from Fig. 5, the mode of the gas power production distribution increases from above 2 TWh/y in 2030 (state “s3_2_up” of node “Gas 2030”) to between 5 and 15 TWh/y in 2050 (state “s3_5_15” of node “Gas 2050”). Correspondingly, the mode of the nuclear power plant production decreases from

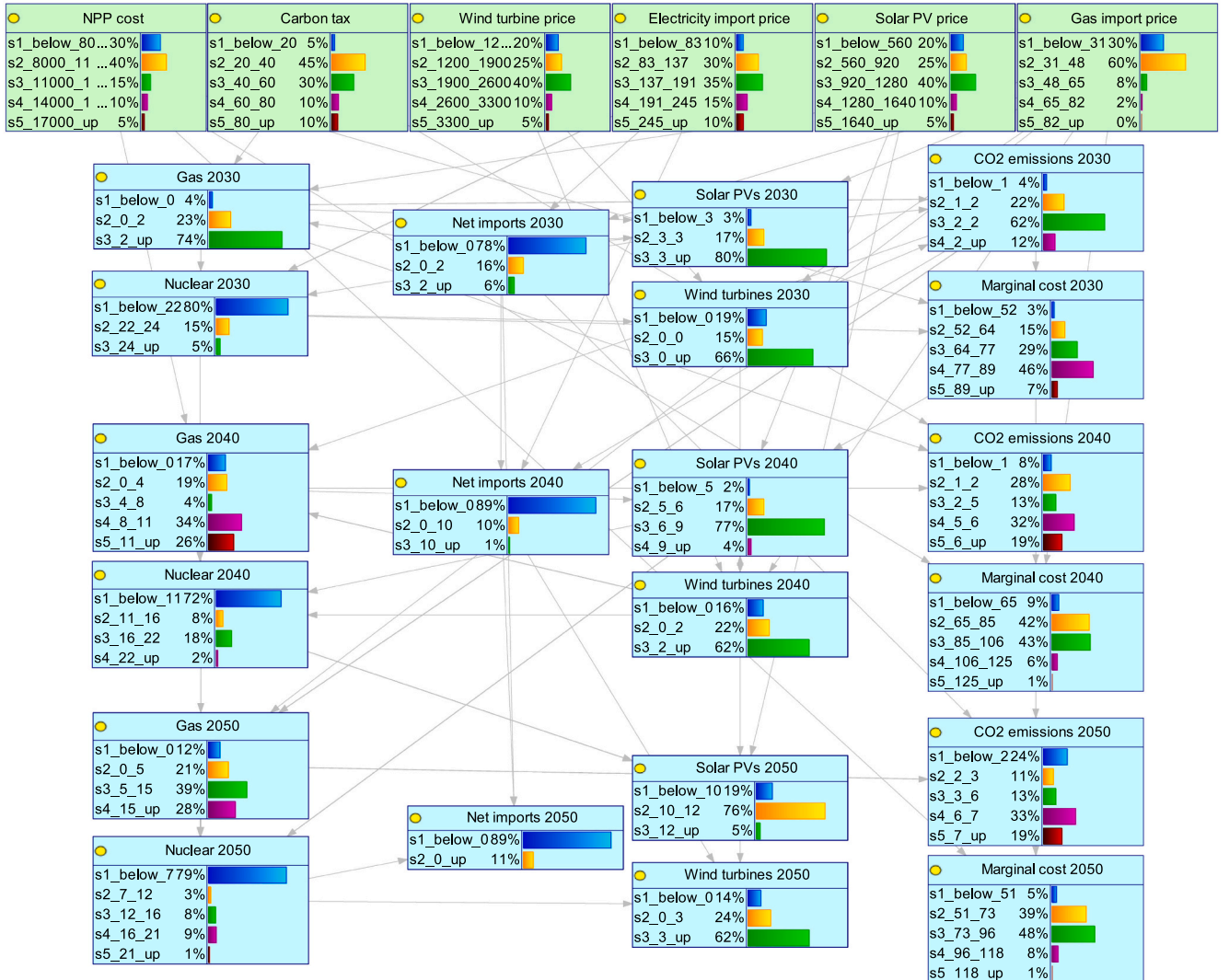


Fig. 5. BBN response on base case scenario. Node definitions and discretization intervals in Appendix B.

below 22 TWh/y in 2030 (produced by the three available Swiss nuclear power plants) to below 7 TWh/y in 2050 (corresponding to about the output of one of the largest Swiss nuclear power plants). Solar PV and wind turbines contribute substantially to the generation capacity throughout all time frames in the base case, with increasing trend of the mode reaching in 2050 between 10 and 12 TWh/h for solar PVs and above 3 TWh/y for wind turbines, which align with expectations from state-of-the-art energy systems analyses [59,60].

3.2. Scenario case 1 – Low emissions

The first scenario case highlights the BBN capability for goal-driven analyses, i.e. backward inference. “Backward” refers to providing evidence on selected BBN output variables and inferring the conditional distributions of the model variables with causal effects associated to the output. For the present energy systems applications, the output evidence represents a policy target and backward, goal-driven, inference allows exploring required changes in the input variables, to meet the set target.

Scenario 1 assigns evidence on the lowest state of “CO₂ emissions 2050”, yellow box in Fig. 6. The figure shows the updated distributions of the BBN nodes, given the evidence, i.e. emissions below 2 Mt./y. To show the causal chains identified by the BBN, the arcs of the parents of the output of interest and of their respective parents are shown in bold in Fig. 6.

In particular, comparing the probability distributions associated to the input variables in the base case and in scenario 1 (nodes on the top

raw of Fig. 5 and in Fig. 6, respectively), the most evident changes involve a decrease in nuclear power production costs (e.g. the probability of the lowest cost state “s1_below_8000” changes from 30 % in the base case to 39 % in scenario 1) and electricity import prices (probability of “s1_below_83” from 10 % to 20 %), with an increase in carbon tax (probability of “s5_80_up” from 9 % to 16 %) and gas import prices (overall shift of the probability mass towards larger price values). These changes favor economic benefits of nuclear power electricity production and imports, which do not contribute to local CO₂ emissions, while discourage the use of fossil fuels, specifically natural gas in the considered STEM-lite realizations.

The primary driver of carbon emissions is electricity production from gas power plants, which in turn is influenced by gas import price and carbon tax (see the bold arrows from “Gas 2050” to “CO₂ emissions 2050” and from “Gas import price” to “Gas 2050” in Fig. 6). Indeed, the BBN model represents the fact that when gas import price increases, electricity generation from gas power plants decreases. This downward shift compared to the base case is observable across all time frames, with a more significant decline starting from 2040. The reduction in electricity production from gas power plants is offset by a larger share of generation from nuclear power plants, wind turbines and increased net imports, visible by comparing the 2050 production shares in Fig. 5 and in Fig. 6.

It is interesting to note that the electricity production prices from wind turbines and solar PVs remain unaffected by the condition to achieve the lowest emissions in 2050 (the corresponding input

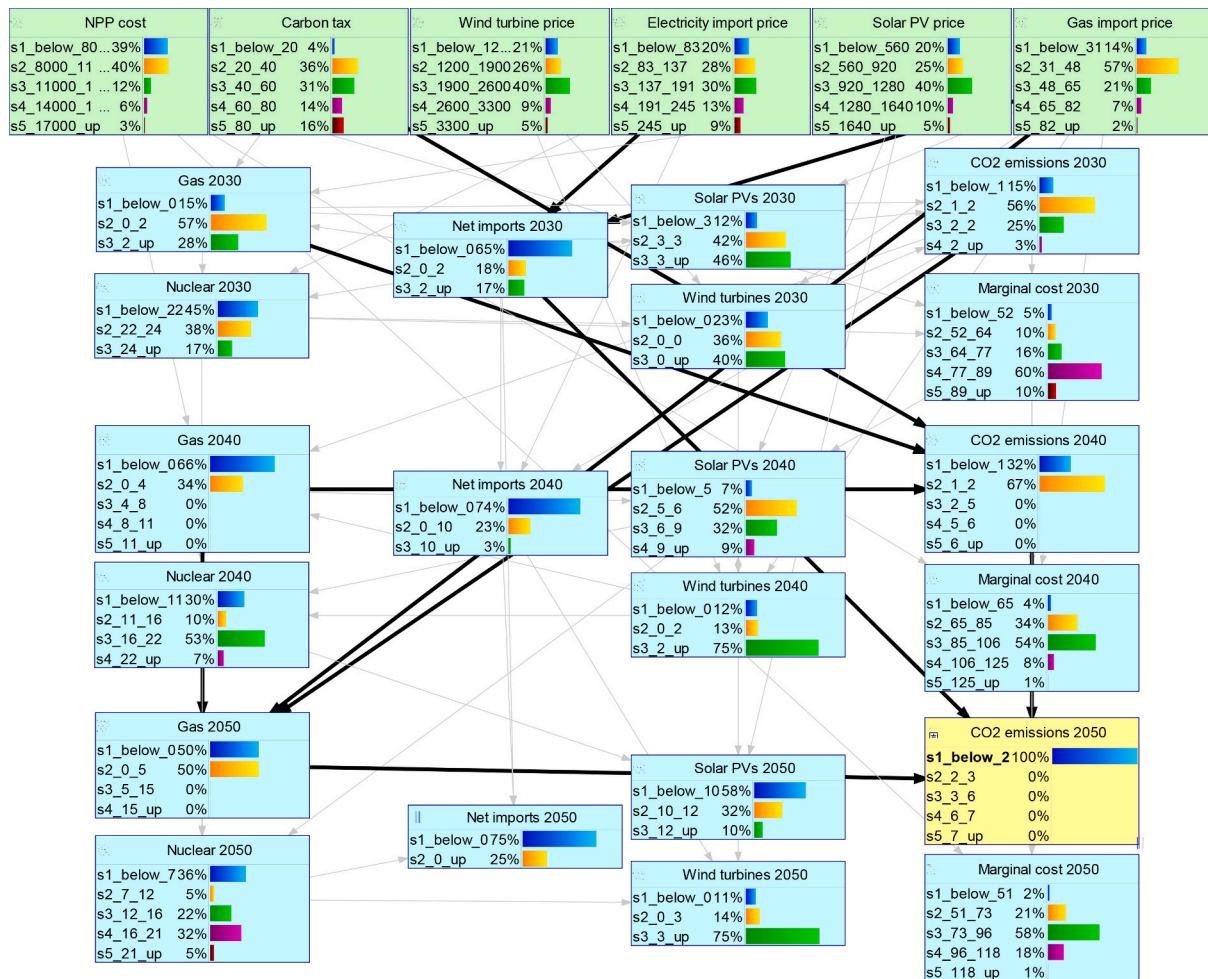
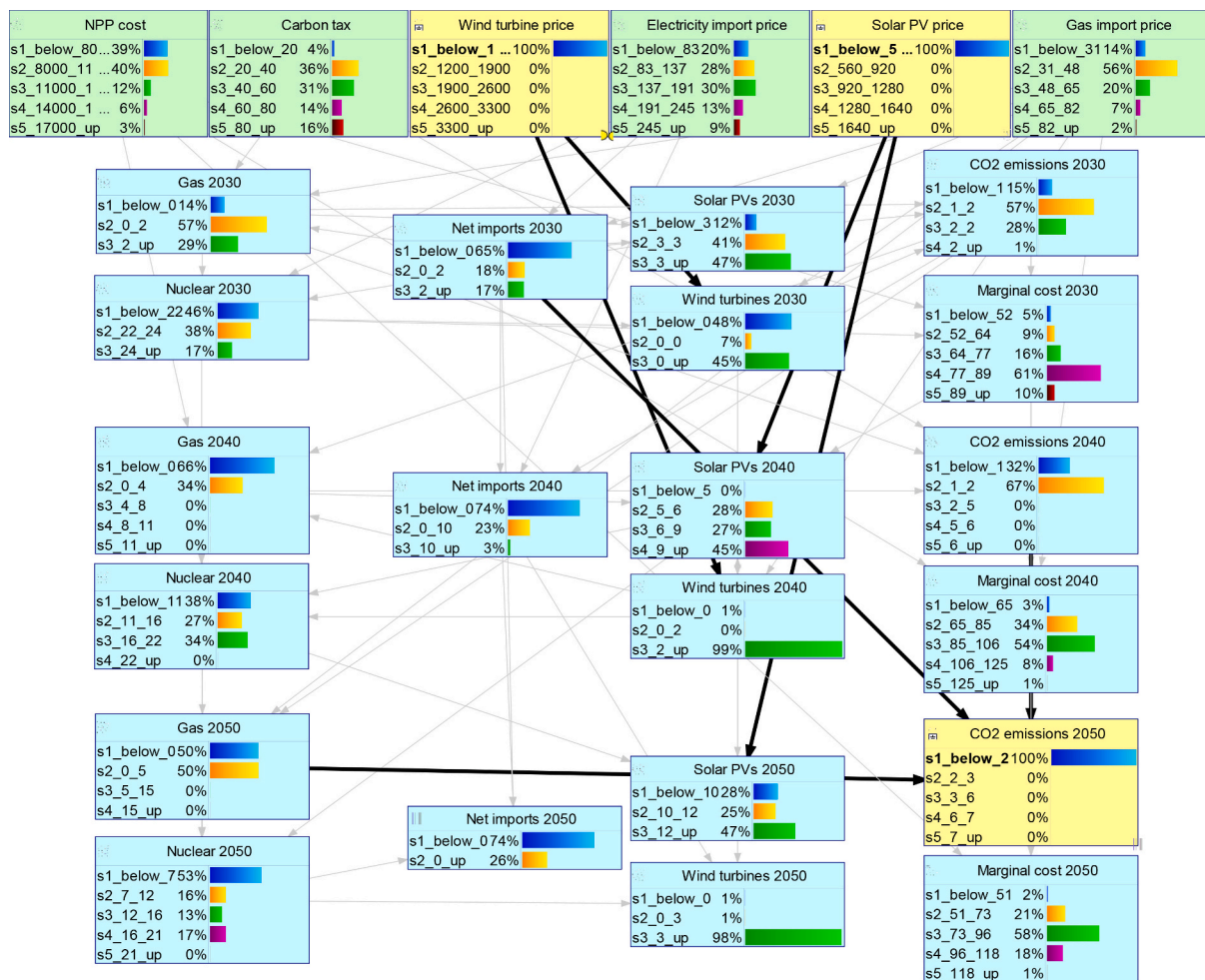


Fig. 6. BBN response in scenario case 1 (arcs of the parents of the output of interest and of their respective parents are shown in bold). Node definitions and discretization intervals in Appendix B. Yellow nodes indicate evidence. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Examining the marginal cost of electricity, we observe a shift upward, although lower prices are still achievable. To achieve the lower

From Fig. 8, the lowest state “s1_below_0” of node “Wind turbines 2030”, results from increased carbon tax values (e.g. probability of state “s5_80_up” changes from 16 % in Fig. 7 to 22 % in Fig. 8 and gas import price (see the mass of the probability distribution shifting downwards)).



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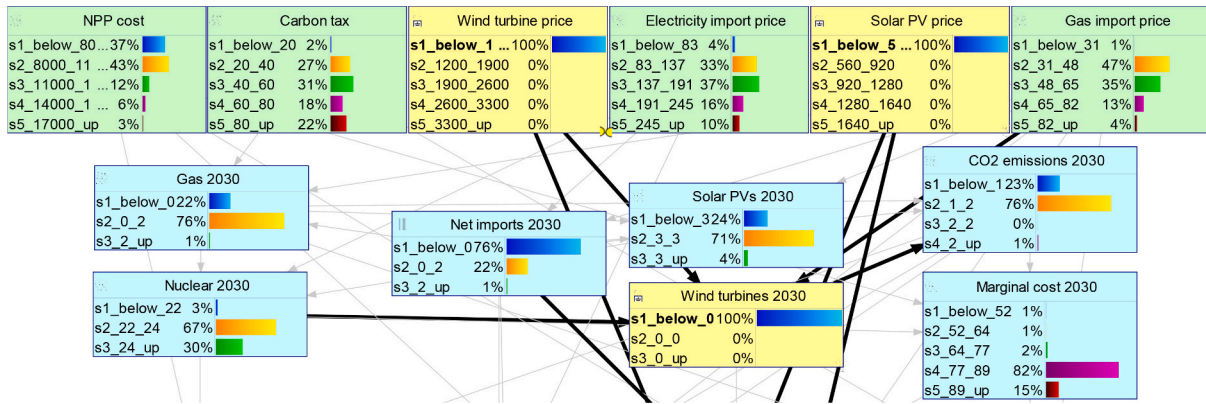


Fig. 8. Scenario case 2, with evidence for node “Wind turbines production 2030” set to its lowest state (“s1_below_0”).

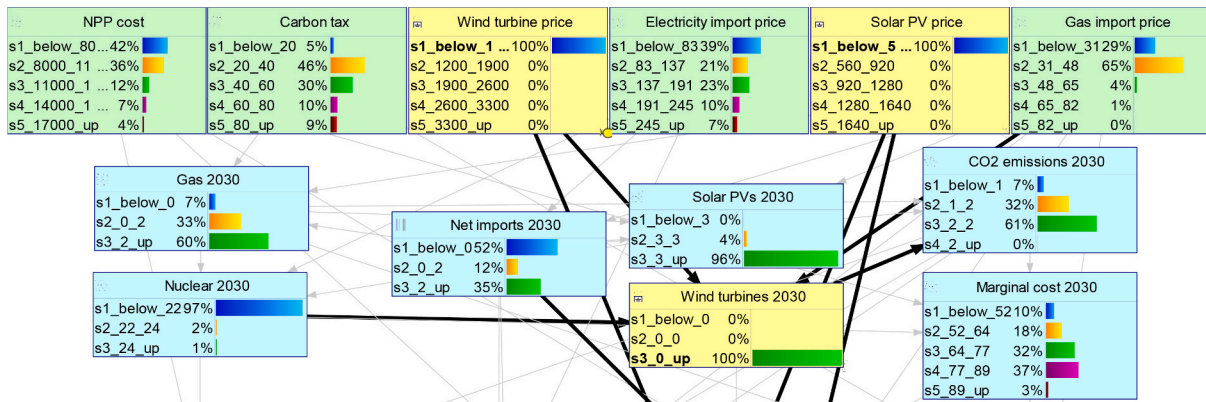


Fig. 9. Scenario case 2, “with evidence for node “Wind turbines production 2030” set to its highest state (“s3_0_up”).

This makes gas power plants less favorable in the energy mix, with increased generation from solar PVs and nuclear power in 2030. From Fig. 9, the highest state of the node “Wind turbines 2030”, “s3_0_up”, corresponds to the electricity production mix produced by lower carbon tax values and gas import prices, which favors larger generation from gas power plants and solar PVs, combined with the decrease from nuclear power generation.

Still with reference to Fig. 8 and Fig. 9, it is interesting to compare the CO₂ emissions and electricity costs associated to the two electricity production mixes in 2030. The mix with smaller gas turbine share (Fig. 8) is associated to lower CO₂ emissions and lower costs, if compared to the mix with largest gas turbine share (Fig. 9).

Going back to the implications of low costs of electricity production for the long-term energy mix, comparing the 2050 electricity shares (scenario cases 1 and 2) shows an increase in the electricity shares by solar PV and wind turbines, as reasonably expected for the analyzed scenario.

4. Discussion and future work

The two scenario cases were designed to demonstrate the BBN application as surrogate energy systems models. As typical for surrogate models, once the model has been developed, BBN models are computationally fast, being based on simple mathematical relationships. An attractive feature for result presentation is the visualization of the node values directly on the model layout. Combined with the (in practice) real-time update of their values, node value visualization well supports user interaction on what-if analyses, exploration of policy options and energy system pathways, possibly during stakeholder demonstrations and decision-making discussion sessions. An example of this application is shown in scenario case 2 demonstrating the effect of low costs of

electricity from renewable sources (e.g. obtained via subsidies) in the long-term electricity production mix.

Still related to visualization of the node values, BBNs offer an intuitive representation of the uncertainty in the inputs and of its propagation to the other model nodes, as shown in Fig. 5, where the visualization of the input marginal distributions was used in support of the definition of the distributions by expert judgment. It is important to note that, while visual representation of the node distributions is attractive for communication and for model scrutiny, for large networks with many nodes it may become difficult for an analyst to understand the drivers for the node uncertainty. Indeed, while sensitivity analysis measures addressing the values and the probabilities of the BBN nodes are integrated in the most common BBN software packages, the application of uncertainty analysis techniques (e.g. variance decomposition-based) is still at pioneering stage, see [61] for a recent application.

A distinguishing feature of BBNs is their support for backward inference (goal-driven inference), i.e. inferring the probability distributions of the influencing factors, given evidence downwards on the causal chain. This was exploited in scenario case 1, where the distributions of the input factors (representing simplifications of policy instruments) allowing for low long-term emissions was inferred. While the case study indeed includes several simplifications, a more comprehensive and realistic BBN can be envisioned for exploration of policies and conditions resulting in the desired energy system performance metrics (e.g. low emissions, low dependence on import, low electricity production costs).

The graphical representation with arrows connecting nodes allows visualizing the node influences as these emerge from the STEM-lite data records, allowing further understanding of the underlying energy systems model relationships. An example was presented in Section 3.2, with reference to Fig. 6. The BBN arrow connections (and lack of connections)

imply that low CO₂ emissions in 2050 are driven by the value of the carbon tax and of the gas import prices, rather than the cost of production from wind turbines and solar PV, if electricity from nuclear power is not decreased (as assumed for the base case, Fig. 5). Another example of the BBN explanatory possibilities is the discussion of the interplay among gas power, wind turbine, and nuclear power electricity production discussed with reference to Fig. 8 and Fig. 9.

Indeed, identification of the correct influences relates to the applicability and performance of BBN structure learning algorithms, itself much influenced by the amount of data available. For the present work, the speed of the STEM-lite allowed the generation of large data sets; BBN models for realistic applications would need to be based on a smaller number of data records, potentially with a larger number of variables. This most likely would imply an increased use of expert judgment in the model structure development and resorting to different types of BBNs. Indeed, labelled node BBNs were a convenient choice for demonstration of the BBN framework features, and future work should envision the adoption of BBN models requiring fewer parameters (e.g. continuous or hybrid BBNs). In particular, continuous BBNs such as Additive Noise Models (ANMs) ([62,63]) and Additive Bayesian Networks (ABNs) [64] represent promising options for future investigation: besides lower data requirements compared to labelled node BBNs, these BBN types offer stronger guarantee for causal identifiability [58]. These advanced BBN implementations have been applied mostly in the context of epidemiological and healthcare studies [64], therefore their adequacy for surrogate modelling will have to be assessed. Concerning the energy systems applications, future research would need to test BBN construction for realistic analysis cases, for example using scenarios generated from the models from [65,66]. Scaling up to realistic applications would involve not only computational aspects (such as dealing with larger networks and fewer data records), but also guidelines to select and define the BBN model nodes, depending on the foreseen application. Indeed, to deal with the computational challenges, one may not necessarily develop a full surrogate model of the energy system, but more specific (and therefore smaller) models each dedicated to address different decision-making issues.

The data sets generated for the purposes of the present paper, along with the more realistic data sets eventually generated from the models from [65,66] will constitute a platform for a systematic performance evaluation of different types of BBNs as well as more traditional surrogate modelling approaches based on machine learning techniques such as Artificial Neural Networks and random forests. It is worth also mentioning that, besides the advantages of continuous over discrete BBNs concerning causality representation and data requirements, the use of continuous BBNs resolves the challenges related to discretizing response [43,44], while however introducing significantly higher computational complexity. These data sets may be also used for a comparison of performance and trade-offs between different continuous and discretization BBN options.

Another important simplification to overcome relates to the treatment of time. For the present application, model inputs were considered constant over the 2030 to 2050 time period, to decrease the size of the BBN model. The variables can still be interpreted as reference values for the whole period and allow deriving meaningful insights as demonstrated in the present paper, however a more informative BBN model should include time dependence on the inputs, e.g. recognizing that policies would not be implemented with constant effects over such a large time frame. Indeed, application of the dynamic BBN variant in energy systems modelling should be subject of future investigation.

5. Conclusion

This paper dealt with the use of BBNs as surrogate models for energy systems applications. A BBN was developed, based on data records generated by a simplified model, STEM-lite. The simplifications introduced in the STEM model and in additional assumptions (e.g. on time

dependence of input variables) were intended to reduce the computation time of single STEM runs and the number of model variables. This allowed working on a BBN based on discrete, labelled node, for which most of the available software packages (GeNIe software in the present work, [46]) offer implemented learning algorithms. Indeed, the aim of the work is to investigate and demonstrate the attractiveness of the modelling framework, eventually leaving the development of algorithmic and computational advances for future work.

The working BBN used for demonstration was developed via an optimization process concerning the number of discretization bin for the input variables, the maximum allowed number of parent nodes, the data set size. For BBN validation, the present work developed BBN performance measures beyond those typically considered in the literature, calculated over 20 % of the generated data records unseen by the BBN in the training phase. In particular, the parameter uniformity measure was developed to address how much the BBN is overparametrized with respect to the available data. Also, validation over the whole distribution of the target variables (referred to as “scenario validation”) was introduced as opposed to validation of only the reference BBN output (e.g. the most likely) for a comprehensive assessment of the BBN performance.

A scenario analysis was designed to demonstrate potential BBN applications, building on a base case and seven scenario cases. The base case and the scenario cases demonstrate three uses of BBNs for energy system modelling: representation of uncertainty in inputs and its propagation through the BBN model, goal-driven analyses, with identification of input values to meet the goals, what-if analyses, exploring effect of changes in input values (of policy implementation) on the model outputs. Indeed, these three uses are not mutually exclusive and can be combined as necessary. The main contribution of this paper is to underscore the BBN attractiveness mainly concerning visualization of the original energy systems model relationships, fast and intuitive uncertainty propagation, backward reasoning. These elements make BBNs a potentially useful tool for energy policy investigation and demonstration, eventually in panel and workshop sessions with experts and diverse stakeholders.

Other potentially attractive features, not exercised in the present work, include BBN's modularity and suitability to represent uncertain effects. Modularity entails the possibility of combining multiple BBNs, each developed from different energy systems models, thus building a progressively more comprehensive surrogate increasing its modelling scope, e.g. adding the transportation sector to the model. Also, the probabilistic nature of BBNs makes these particularly suitable to modelling of “soft influences”, e.g. those related human factors. For energy systems applications, this could be useful to represent uncertainty in the adoption of policies, consumer behaviors and socio-economical influences.

Building on the attractive features, exploiting BBNs for practical purposes in energy systems modelling requires overcoming a number of simplifying assumptions made in the present work. In particular, this would involve dealing with smaller sets of data records to train the BBN, allow for time dependence in the input variables, and generally dealing with larger BBN models (i.e. involving more variables). The identified directions of future research work involve adoption of BBN variants with smaller number of parameters (e.g. continuous or hybrid), incorporating dynamics, more extensive use of expert judgment to develop the BBN structure.

CRediT authorship contribution statement

Rainer Kelk: Writing – original draft, Methodology, Investigation, Conceptualization. **Luca Podofillini:** Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Vinh N. Dang:** Writing – review & editing, Supervision. **Evangelos Panos:** Writing – review & editing, Supervision, Investigation, Conceptualization.

Disclaimer

The views expressed in this work are solely those of the authors. The authors bear sole responsibility for the conclusions and the results presented in this publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Appendix A. Fundamentals of Bayesian Belief Networks

A Bayesian network consists of a set of N variables, $X_1, \dots, X_N$ ($i = 1, 2, \dots, N$), and a set of directed edges between them [16]. For discrete BBNs, each variable, X_i , can assume any of n_i mutually exclusive states, with values $x_{i,1}, x_{i,2}, \dots, x_{i,n_i}$. The variables and the directed edges form a Directed Acyclic Graph (DAG): “acyclic” entails that for every X_i there is no path returning to X_i . Each edge connects a “parent” node to a “child” node. A set of conditional probability distributions, $P(X_i|pa(X_i))$, is associated to each variable X_i with parents $pa(X_i)$, one distribution for each combination of the parent states. For variables without parents, the distribution reduces to the unconditional distribution, e.g. $P(X_i)$.

A BBN provides a compact and visual representation of the joint probability distribution $P(X_1, \dots, X_N)$ of the N variables $X_1, \dots, X_N$. The DAG drives the factorization of $P(X_1, \dots, X_N)$ according to:

$$P(X_1, \dots, X_N) = \prod_{i=1}^N P(X_i|pa(X_i)) \quad (\text{A.1})$$

where $pa(X_i)$ is the set of parents of X_i as represented by the DAG.

As an example, Fig. 10 shows a BBN involving $N = 5$ variables, $X_1, X_2, \dots, X_5$. As guided by the DAG relationships, the joint probability distribution of $X_1, X_2, \dots, X_5$ can be written as:

$$P(X_1, X_2, X_3, X_4, X_5) = P(X_1)P(X_2)P(X_3|X_1, X_2)P(X_4|X_1, X_3)P(X_5|X_3) \quad (\text{A.2})$$

For example, $P(X_3|X_1, X_2)$ contains $n_1 \bullet n_2$ conditional probability distributions, each of the form $P(X_3|x_{1,j}, x_{2,k})$, with $j = 1, 2, \dots, n_1$ and $k = 1, 2, \dots, n_2$.

The BBN formalism allows probability updating given evidence, E , typically in the form of observed values for the variables, eg. $X_5 = x_{5,2}$ and $X_2 = x_{2,3}$ or knowledge of their probability (either from data or judgment), e.g. $P(X_5) = (p_{5,1}, p_{5,2}, \dots, p_{5,n_5})$.

Eq. A.1 is the basis to propagate the evidence E and update the joint probability to $P(X_1, \dots, X_N|E)$. Evidence can be set to any of the BBN variables. Often, if set at the top of the causal chain, evidence propagation is referred to as forward inference, while if at the bottom it is referred to as backward inference. Probability updating is based on the fundamental Bayes’ rule (hence the name of the modelling technique), which, expressed for two variables A and B , reads:

$$P(A|B) = \frac{P(B|A)}{P(B)} \quad (\text{A.3})$$

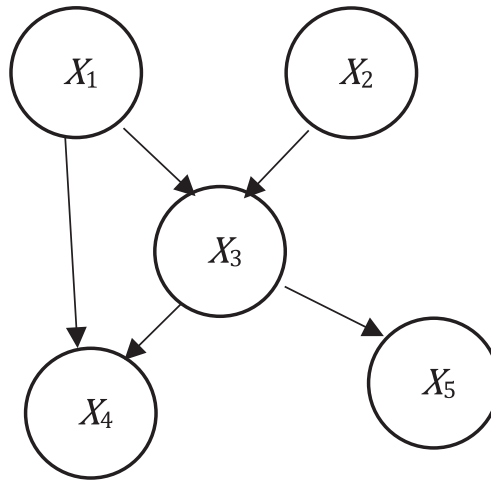


Fig. 10. Example BBN.

Appendix B. BBN node definitions and discretization intervals

Node name	Description (unit of continuous variable)	State label	Lower bound	Upper bound
Gas 2030	Gas power plant production 2030 (TWh/yr)	s1_below_0	0.14	0.23
		s2_0_2	0.23	2.00
		s3_2_up	2.00	3.59
Nuclear 2030	Nuclear power plant production 2030 (TWh/yr)	s1_below_22	20.24	22.00
		s2_22_24	22.00	24.30
		s3_24_up	24.30	25.13
Net imports 2030	Net imports 2030 (TWh/yr)	s1_below_0	0.00	0.00
		s2_0_2	0.00	2.00
		s3_2_up	2.00	3.59
Solar PV 2030	Solar PV production 2030 (TWh/yr)	s1_below_3	2.83	3.28
		s2_3_3	3.28	3.75
		s3_3_up	3.75	4.06
Wind turbine 2030	Wind turbine production 2030 (TWh/yr)	s1_below_0	0.12	0.27
		s2_0_0	0.27	0.57
		s3_0_up	0.57	1.07
CO2 emissions 2030	CO2 emissions 2030 (Mt/yr)	s1_below_1	1.34	1.40
		s2_1_2	1.40	2.00
		s3_2_2	2.00	2.90
Marginal cost 2030 (CHF/MWh)	Marginal cost of electricity 2030 (CHF/MWh)	s4_2_up	2.90	3.08
		s1_below_52	39.73	52.00
		s2_52_64	52.00	64.00
Gas 2040	Gas power plant production 2040 (TWh/yr)	s3_64_77	64.00	77.00
		s4_77_89	77.00	89.00
		s4_89_up	89.00	102.23
Nuclear 2040	Nuclear power plant production 2040 (TWh/yr)	s1_below_0	0.00	0.10
		s2_0_4	0.10	4.83
		s3_4_8	4.83	8.77
Net imports 2040	Net imports 2040 (TWh/yr)	s4_8_11	8.77	11.96
		s5_11_up	11.96	16.43
		s1_below_11	6.58	11.24
Solar PV 2040	Solar PV production 2040 (TWh/yr)	s2_11_16	11.24	16.19
		s3_16_22	16.19	22.66
		s4_22_up	22.66	25.54
Wind turbine 2040	Wind turbine production 2040 (TWh/yr)	s1_below_0	0.00	0.20
		s2_0_10	0.20	10.00
		s3_10_up	10.00	13.11
CO2 emissions 2040	CO2 emissions 2040 (Mt/yr)	s1_below_5	4.32	5.34
		s2_5_6	5.34	6.77
		s3_6_9	6.77	9.98
Marginal cost 2040	Marginal cost of electricity 2040 (CHF/MWh)	s4_9_up	9.98	16.43
		s1_below_0	0.14	0.66
		s2_0_2	0.66	2.00
Gas 2050	Gas power plant production 2050 (TWh/yr)	s3_2_up	2.00	2.69
		s1_below_1	0.38	1.00
		s2_1_2	1.00	2.98
Nuclear 2050	Nuclear power plant production 2050 (TWh/yr)	s3_2_5	2.98	5.50
		s4_5_6	5.50	6.40
		s5_6_up	6.40	8.29
Net imports 2050	Net imports 2050 (TWh/yr)	s1_below_65	32.51	65.00
		s2_65_85	65.00	85.00
		s3_85_106	85.00	106.00
Solar PV 2050	Solar PV production 2050 (TWh/yr)	s4_106_125	106.00	125.00
		s5_125_up	125.00	215.97
		s1_below_0	0.00	0.00
Wind turbine 2050	Wind turbine production 2050 (TWh/yr)	s2_0_5	0.00	5.00
		s3_5_15	5.00	15.00
		s4_15_up	15.00	19.00
CO2 emissions 2050	CO2 emissions 2050 (Mt/yr)	s1_below_7	1.25	7.00
		s2_7_12	7.00	12.00
		s3_12_16	12.00	16.00
Marginal cost 2050	Marginal cost of electricity 2050 (CHF/MWh)	s4_16_21	16.00	21.00
		s6_21_up	21.00	23.35
		s1_below_0	0.00	0.00
Gas 2030	Gas power plant production 2030 (TWh/yr)	s2_0_up	0.00	17.00
		s1_below_10	7.85	10.35
		s2_10_12	10.35	12.00
Nuclear 2030	Nuclear power plant production 2030 (TWh/yr)	s3_12_up	12.00	24.00
		s1_below_0	0.18	0.50
		s2_0_3	0.50	3.00
Net imports 2030	Net imports 2030 (TWh/yr)	s3_3_up	3.00	4.30
		s1_below_2	0.37	2.00

(continued on next page)

(continued)

Node name	Description (unit of continuous variable)	State label	Lower bound	Upper bound
Marginal cost of 2050	Marginal cost of electricity 2050 (CHF/MWh)	s2_2_3	2.00	3.50
		s3_3_6	3.50	6.00
		s4_6_7	6.00	7.50
		s5_7_up	7.50	8.30
		s1_below_51	29.01	51.00
		s2_51_73	51.00	73.00
		s3_73_96	73.00	96.00
		s4_96_118	96.00	118.00
		s4_118_up	118.00	140.90

Appendix C. BBN scenario cases

This appendix includes the BBN responses for scenario cases 3 to 7 from Table 2. Fig. 11, Scenario case 3 from Table 2 - Low emissions.

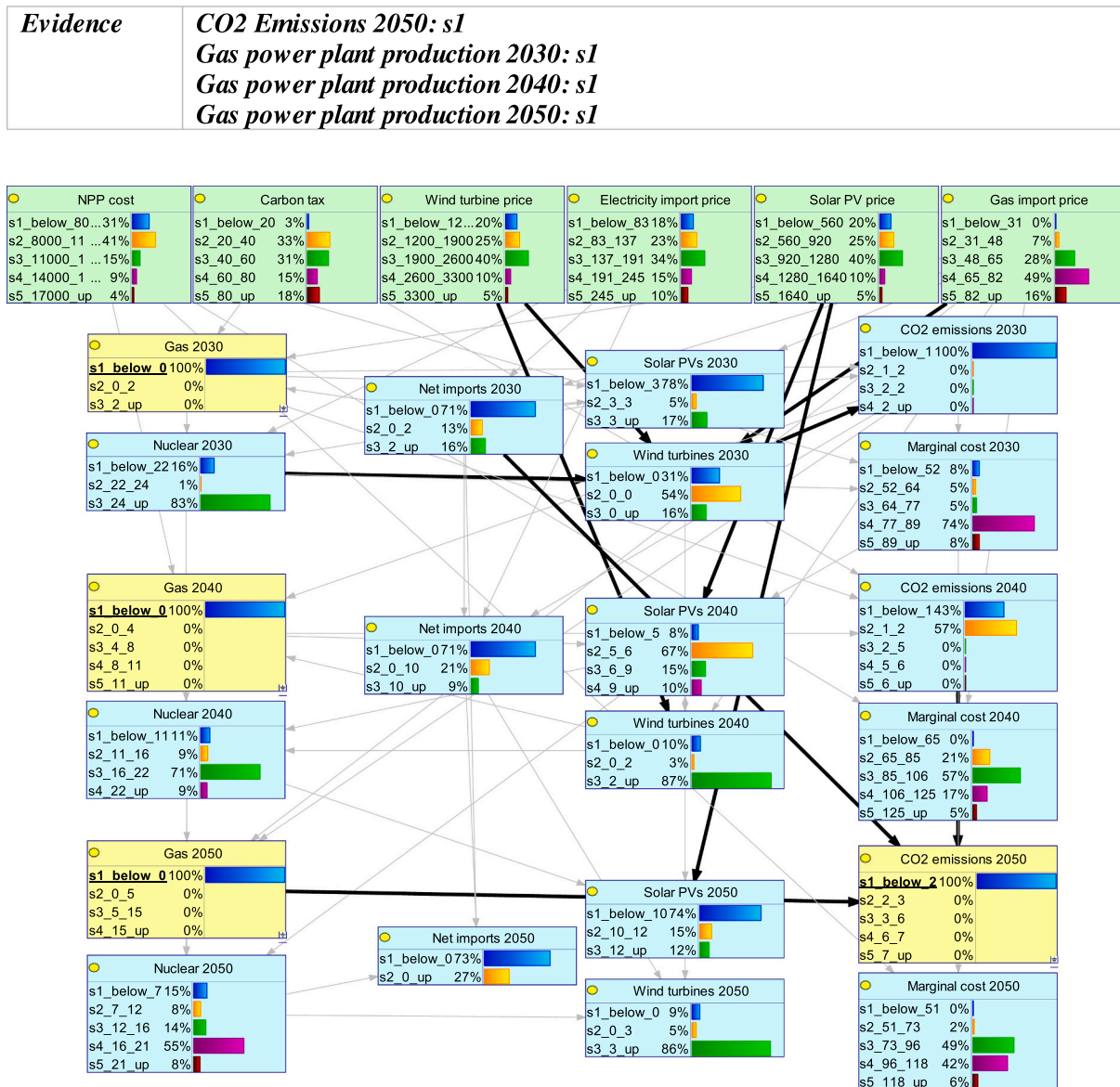


Fig. 11. BBN response on Scenario case 3, Low emissions, no gas. Yellow nodes indicate evidence. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fig. 12. Scenario case 4 from Table 2 - Low emissions, low renewable prices.

Evidence	<i>CO2 Emissions 2050: s1</i> <i>Wind turbine prices: s1</i> <i>Solar PV prices: s1</i> <i>Gas power plant production 2030: s1</i> <i>Gas power plant production 2040: s1</i> <i>Gas power plant production 2050: s1</i>
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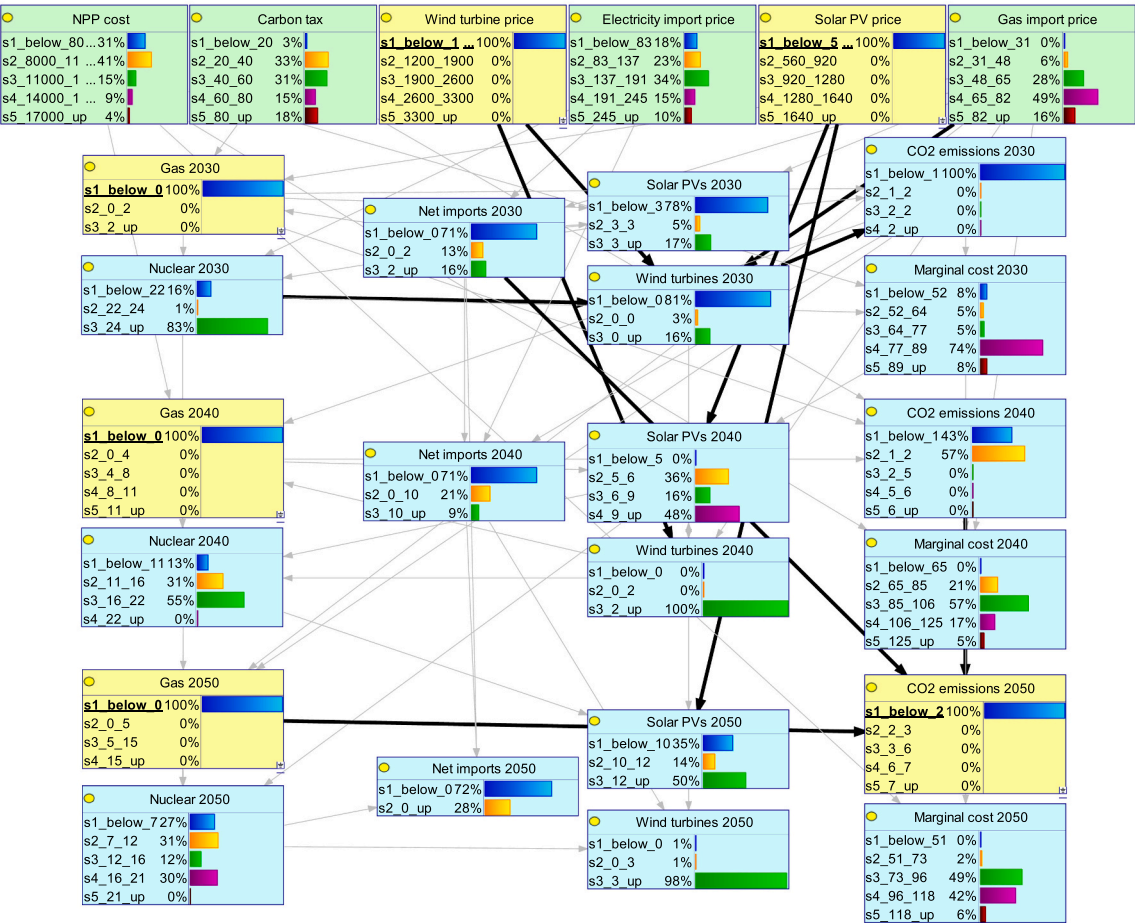


Fig. 12. BBN response on Scenario case 4, Low emissions, low renewable prices, no gas. Yellow nodes indicate evidence. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fig. 13, Scenario case 5 from Table 2 - Low emissions, low renewable prices.

<i>Evidence</i>	<i>CO2 Emissions 2050: s1</i> <i>Wind turbine prices: s1</i> <i>Solar PV prices: s1</i> <i>Net import 2030: s1</i> <i>Net import 2040: s1</i> <i>Net import 2050: s1</i>
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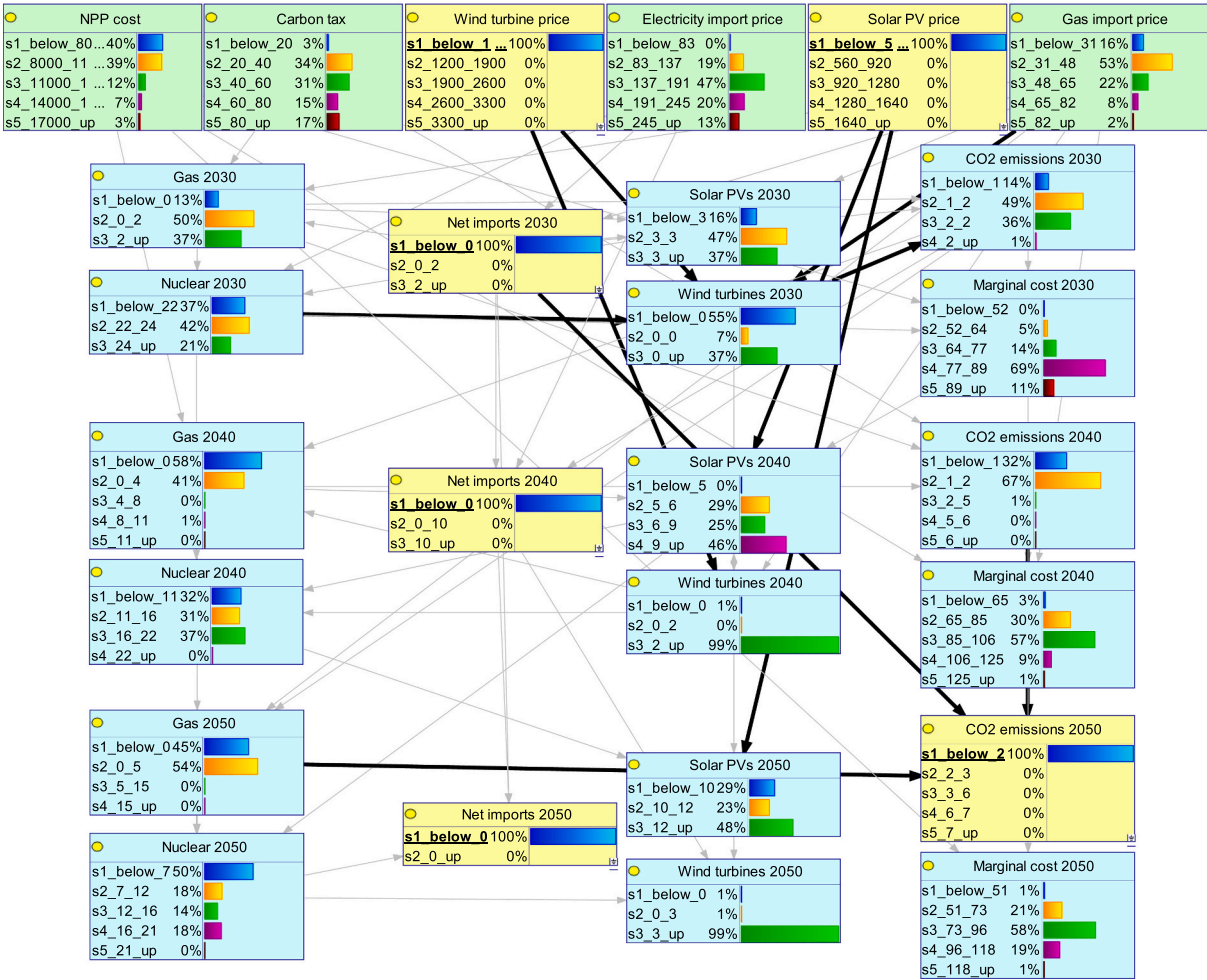


Fig. 13. BBN response on Scenario case 5, Low emissions, low renewable prices, no imports. Yellow nodes indicate evidence. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fig. 14. Scenario case 6 from Table 2 - Low emissions, low renewable prices, no imports.

Evidence	<i>CO2 Emissions 2050: s1</i> <i>Wind turbine prices: s1</i> <i>Solar PV prices: s1</i> <i>Net import 2030: s1</i> <i>Net import 2040: s1</i> <i>Net import 2050: s1</i> <i>Gas power plant production 2030: s1</i> <i>Gas power plant production 2040: s1</i> <i>Gas power plant production 2050: s1</i>
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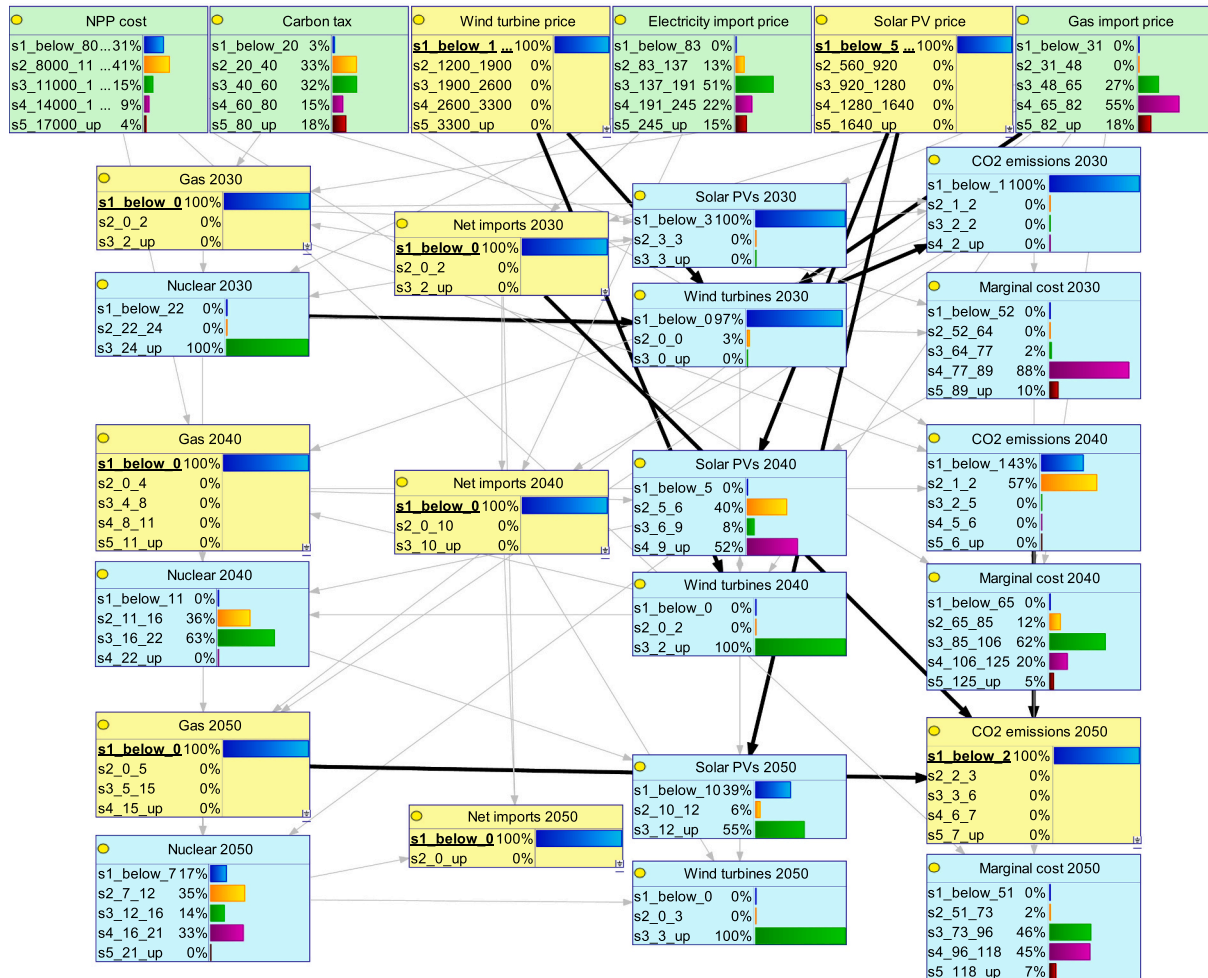


Fig. 14. BBN response on Scenario case 6, Low emissions, low renewable prices, no imports, no gas. Yellow nodes indicate evidence. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fig. 15. Scenario case 7 from Table 2 - Low emissions, high electricity and gas.

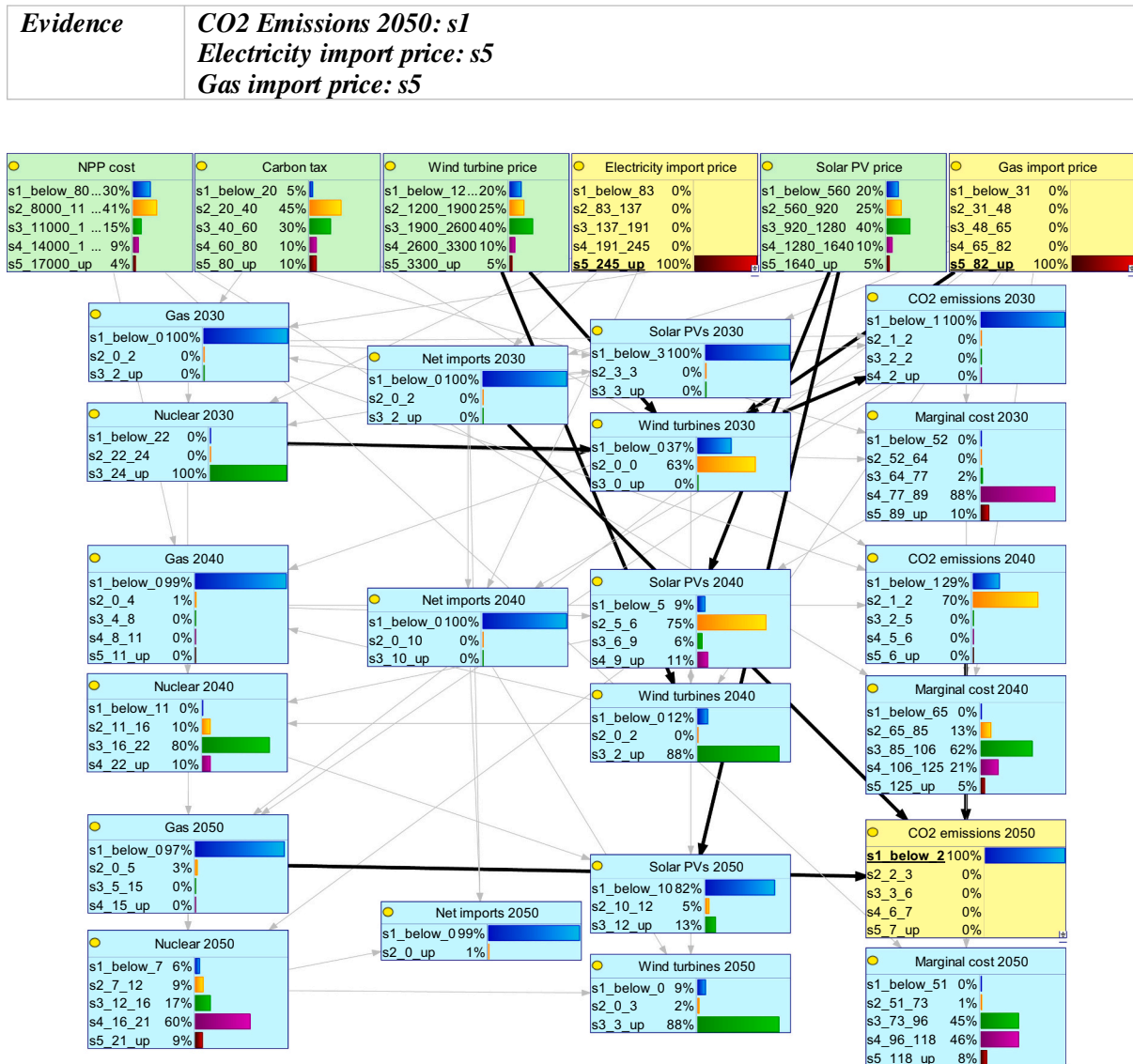


Fig. 15. BBN response on Scenario case 7, Low emissions, high electricity and gas prices. Yellow nodes indicate evidence. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Data availability

Data will be made available on request.

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